LRU:

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     LRU
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     54231
  - Cache Hits:         38962
  - Cache Misses:       15269
  - Hit Rate:           71.85%
  - Miss Rate:          28.15%
  - IPC:                0.208
  - Total Cycles:       481725
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:   1
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Direct Mapped
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     LRU
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     52847
  - Cache Hits:         42278
  - Cache Misses:       10569
  - Hit Rate:           79.99%
  - Miss Rate:          20.01%
  - IPC:                0.275
  - Total Cycles:       364225
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            32
  - L1 Associativity:   2
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               2-way Set Associative
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     LRU
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     51923
  - Cache Hits:         44347
  - Cache Misses:       7576
  - Hit Rate:           85.41%
  - Miss Rate:          14.59%
  - IPC:                0.346
  - Total Cycles:       289400
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            16
  - L1 Associativity:   4
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               4-way Set Associative
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     LRU
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     51284
  - Cache Hits:         46026
  - Cache Misses:       5258
  - Hit Rate:           89.75%
  - Miss Rate:          10.25%
  - IPC:                0.432
  - Total Cycles:       231450
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Fully Associative
==============================================================
```

FIFO:

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     FIFO
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     55187
  - Cache Hits:         37827
  - Cache Misses:       17360
  - Hit Rate:           68.54%
  - Miss Rate:          31.46%
  - IPC:                0.187
  - Total Cycles:       534500
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 true
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:   1
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Direct Mapped
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     FIFO
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     53456
  - Cache Hits:         41346
  - Cache Misses:       12110
  - Hit Rate:           77.36%
  - Miss Rate:          22.64%
  - IPC:                0.248
  - Total Cycles:       402750
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 true
Cache Configuration from Knobs:
  - L1 Sets:            32
  - L1 Associativity:   2
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               2-way Set Associative
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     FIFO
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     52198
  - Cache Hits:         44511
  - Cache Misses:       7687
  - Hit Rate:           85.27 %
  - Miss Rate:          14.73 %
  - IPC:                0.342
  - Total Cycles:       292175
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 true
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Fully Associative
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     FIFO
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     52735
  - Cache Hits:         43062
  - Cache Misses:       9673
  - Hit Rate:           81.65 %
  - Miss Rate:          18.35 %
  - IPC:                0.248
  - Total Cycles:       341825
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 true
Cache Configuration from Knobs:
  - L1 Sets:            16
  - L1 Associativity:   4
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               4-way Set Associative
==============================================================
```

CLOCK:

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     CLOCK
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     54709
  - Cache Hits:         38394
  - Cache Misses:       16315
  - Hit Rate:           70.19 %
  - Miss Rate:          29.81 %
  - IPC:                0.197
  - Total Cycles:       507875
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:   1
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Direct Mapped
==============================================================
```

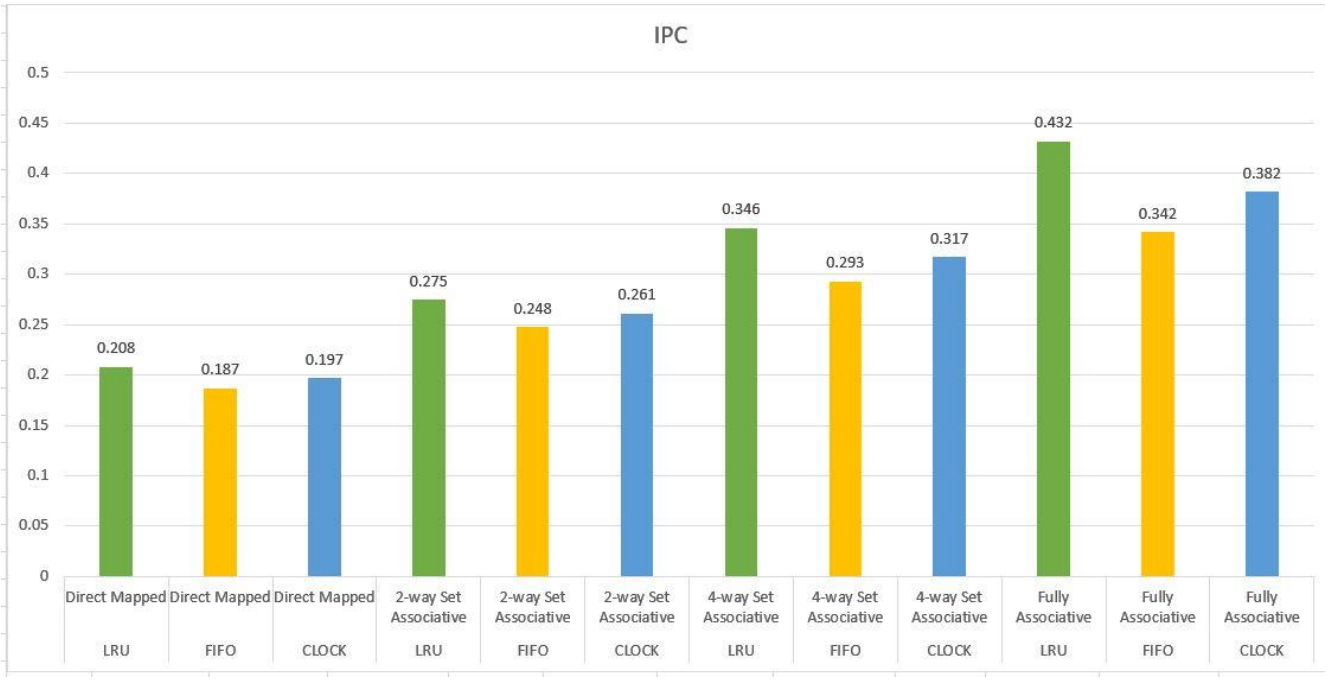
```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     CLOCK
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     53152
  - Cache Hits:         41812
  - Cache Misses:       11340
  - Hit Rate:           78.66 %
  - Miss Rate:          21.34 %
  - IPC:                0.261
  - Total Cycles:       383500
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            32
  - L1 Associativity:   2
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               2-way Set Associative
==============================================================
```

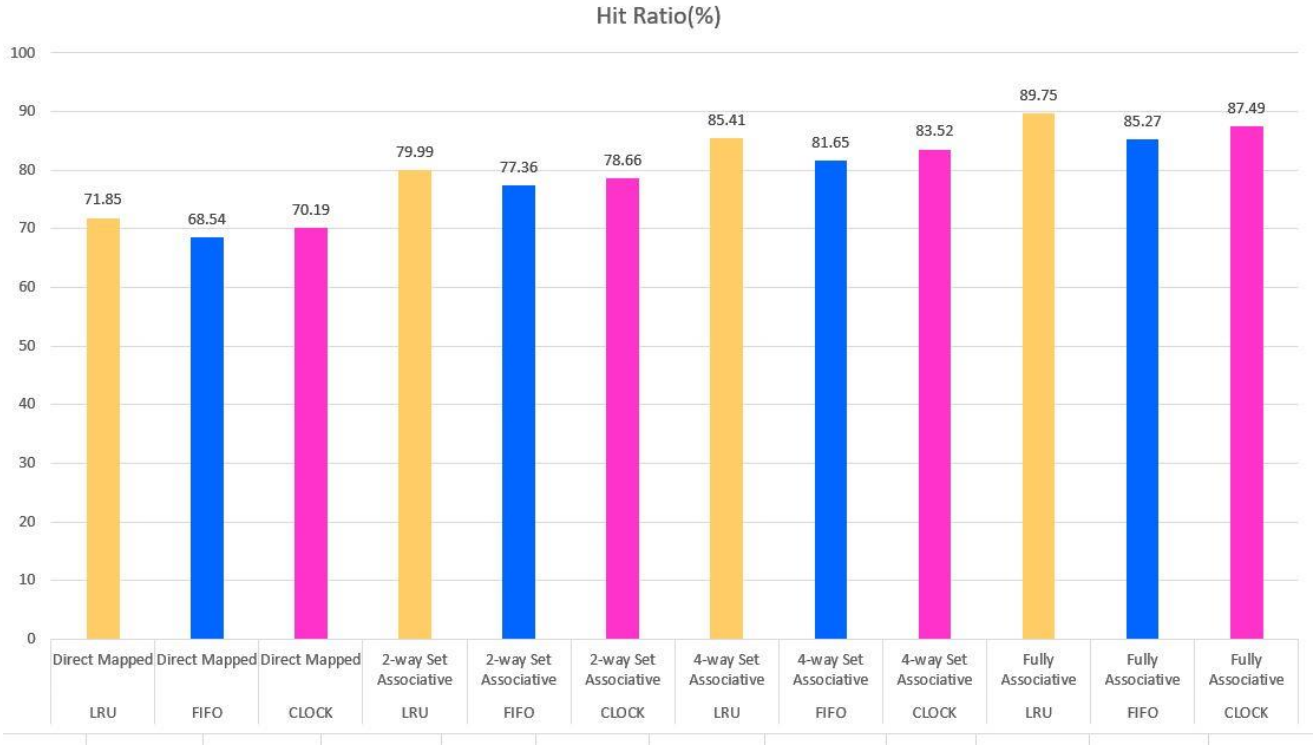
```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     CLOCK
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     52329
  - Cache Hits:         43705
  - Cache Misses:       8624
  - Hit Rate:           83.52 %
  - Miss Rate:          16.48 %
  - IPC:                0.317
  - Total Cycles:       315600
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            16
  - L1 Associativity:   4
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               4-way Set Associative
==============================================================
```

```
src/macsim.cc:844: (I=0  C=0):  NOC clock frequency : 1.6 GH
src/macsim.cc:845: (I=0  C=0):  MC  clock frequency : 2.1 GH
src/process_manager.cc:455: (I=0  C=0):  setup_process:0 /ho
src/process_manager.cc:548: (I=0  C=0):  thread_count:1
**Core 00 Thread 00 Finished:   insts:100000      cycles:423
src/retire.cc:597: (I=99999  C=11857132):  ----- application
2
**Core 00 Core_Total  Finished:   insts:100000      cycles:4
==============================================================
                        L1D CACHE SIMULATION RESULTS
==============================================================
Replacement Policy:     CLOCK
Cache Configuration:
  - Cache Type:         L1 Data Cache
  - Total Cache Size:   Available via knobs
--------------------------------------------------------------
Performance Statistics:
  - Total Accesses:     51741
  - Cache Hits:         45269
  - Cache Misses:       6472
  - Hit Rate:           87.49 %
  - Miss Rate:          12.51 %
  - IPC:                0.382
  - Total Cycles:       261800
  - Instructions (C0):  100000
--------------------------------------------------------------
Knob Settings Used:
  - cache_use_pseudo_lru:    false
  - wb_fifo:                 false
Cache Configuration from Knobs:
  - L1 Sets:            64
  - L1 Associativity:
  - L1 Line Size:       128 bytes
  - Total Cache Size:   8192 bytes
  - Type:               Fully Associative
==============================================================
```

GRAPH

Replacement Policy	LRU	LRU	LRU	LRU	FIFO	FIFO	FIFO	FIFO	CLOCK	CLOCK	CLOCK	CLOCK
Configuration	Direct Mapped	2-way Set Associative	4-way Set Associative	Fully Associative	Direct Mapped	2-way Set Associative	4-way Set Associative	Fully Associative	Direct Mapped	2-way Set Associative	4-way Set Associative	Fully Associative
Total Accesses	54,231	52,847	51,923	51,284	55,187	53,456	52,735	52,198	54,709	53,152	52,329	51,741
Cache Hits	38,962	42,278	44,347	46,026	37,827	41,346	43,062	44,511	38,394	41,812	43,705	45,269
Cache Misses	15,269	10,569	7,576	5,258	17,360	12,110	9,673	7,687	16,315	11,340	8,624	6,472
Hit Ratio(%)	71.85	79.99	85.41	89.75	68.54	77.36	81.65	85.27	70.19	78.66	83.52	87.49
Miss Ratio(%)	28.15	20.01	14.59	10.25	31.46	22.64	18.35	14.73	29.81	21.34	16.48	12.51
Total Cycles	481,725	364,225	289,400	231,450	534,500	402,750	341,825	292,175	507,875	383,500	315,600	261,800
IPC	0.208	0.275	0.346	0.432	0.187	0.248	0.293	0.342	0.197	0.261	0.317	0.382





- Going from direct mapped to 2-way set associative gives big improvements. For example, with LRU: Hit ratio goes from 71.85% to 79.99% (up 8.14 points), IPC goes from 0.208 to 0.275 (up 0.067), Cache misses drop from 15,269 to 10,569 (down 4,700 misses) Going from 2-way to 4-way still helps, but not as much. The hit ratio only goes up 5.42 points and IPC increases by 0.071. We can refer that adding more ways helps a lot.
- LRU consistently beats the other replacement policies. Looking at 4-way set associative caches: LRU: 85.41% hit ratio, 0.346 IPC; CLOCK: 83.52% hit ratio, 0.317 IPC; FIFO: 81.65% hit ratio, 0.293 IPC. LRU beats FIFO by 3.76 points in hit ratio and 0.053 in IPC. CLOCK sits in the middle, performing 1.89 points better than FIFO in hit ratio. So, LRU works the best, FIFO works worst.
- With total accesses better caches actually make fewer total memory requests: FIFO direct mapped: 55,187 total accesses; CLOCK direct mapped: 54,709 total accesses; LRU direct mapped: 54,231 total accesses. LRU needs 956 fewer accesses than FIFO even for the same workload. This happens because better caches reduce pipeline stalls. So, better caches need fewer memory accesses.
- Every cache miss costs 25 extra cycles. This makes hit ratios very important. When comparing two extremes: FIFO direct mapped: 17,360 misses = 434,000 extra cycles; LRU fully associative: 5,258 misses = 131,450 extra cycles. The LRU cache saves 302,550 cycles just by avoiding misses. That's why the IPC difference is so large, as the 25-cycle miss penalty hurts performance.
- Direct mapped caches should perform the same regardless of replacement policy since there's only one place to put data. But our results show differences: LRU direct mapped: 71.85% hit ratio; CLOCK direct mapped: 70.19% hit ratio; FIFO direct mapped: 68.54% hit ratio. There's a 3.31 point difference between best and worst. This might be due to how the cache warms up or interacts with the processor pipeline.
- Going from 4-way to fully associative gives smaller improvements: LRU: Hit ratio goes from 85.41% to 89.75% up 4.34 points, IPC goes from 0.346 to 0.432 up 0.086. While this is still good, it's much smaller than the 8.14 point improvement from direct mapped to 2-way. The hardware cost of fully associative caches is very high for this smaller gain. So, these improvements stop being worth it.

For high performance: By using 4-way set associative with LRU, we will get 85.41% hit ratio and 0.346 IPC without the complexity of fully associative.

For balanced design: By using 4-way set associative with CLOCK, we will get 83.52% hit ratio and 0.317 IPC with simpler hardware than LRU.

For simple designs: By using at least 2-way set associative with CLOCK, we will get 78.66% hit ratio and 0.261 IPC. We don't need to use direct mapped unless have to.

- Cache performance affects everything the processor does. A cache that improves IPC from 0.187 to 0.346 makes programs run almost twice as fast. This is why modern processors spend a lot of effort on cache design. As we can see that the cycle count differences are huge. LRU fully associative uses 231,450 cycles while FIFO direct mapped uses 534,500 cycles. That's 303,050 more cycles for the same work.
- Based on our results, 4-way set associative cache is the best practical choice as, it is balanced and gives most of the performance benefits without extreme hardware complexity. The jump from 2-way to 4-way is still good, but going beyond 4-way gives much smaller improvements.
- For replacement policies, LRU is best when you can afford the complexity. CLOCK is a middle choice that's much simpler to build but still performs well. FIFO should be avoided unless hardware resources are extremely limited. However, the best configuration is LRU fully associative. As it has an IPC of 0.432 while the worst is FIFO direct mapped, as it has 0.187. This means the best cache is 2.3 times faster than the worst cache.

Ece Düzgeç
20220702045